

Data Communications and Networking

Project 2

Wireshark Lab: TCP

Answer the following questions, from your captured trace.

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu? (Hint: To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window").

Answer:

The screenshot shows a Wireshark capture of an HTTP transaction. The packet list pane at the top shows two packets: a POST request (No. 261) and an OK response (No. 281). The selected packet (No. 261) is expanded in the packet details pane, showing the Hypertext Transfer Protocol section with the method POST and the file name alice.txt. The packet bytes pane at the bottom shows the raw data of the request, including the file content.

No.	Time	Source	Destination	Protocol	Length	Info
261	3.940112	192.168.1.222	128.119.245.12	HTTP	41415	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
281	4.990853	128.119.245.12	192.168.1.222	HTTP	831	HTTP/1.1 200 OK (text/html)

Details of selected packet (No. 261):

- Frame 261: 41415 bytes on wire (331320 bits), 41415 bytes captured (331320 bits) on interface \Dev...
- Ethernet II, Src: Intel_4f:f8:bd (48:51:c5:4f:f8:bd), Dst: Sageneadboa_29:fc:92 (f0:7b:65:29:fc:92)
- Internet Protocol Version 4, Src: 192.168.1.222, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 53670, Dst Port: 80, Seq: 111573, Ack: 1, Len: 41361
- Hypertext Transfer Protocol, Method: POST, URI: /wireshark-labs/lab3-1-reply.htm, Version: 1.1, Content-Type: text/plain, Content-Length: 41361

Packet bytes (Frame 261):

```
0000 04 02 30 11 00 00 89 64 20 41 6e 69 63 65 26 0e ...0...did Alice...
0010 0a 0d 0a 20 20 68 57 68 79 2c 20 53 48 45 2c 27 ...why, she...
0020 20 73 61 69 64 20 74 68 65 20 47 72 79 70 68 6f ...said the Grypho...
0030 6e 2e 20 20 68 49 74 27 73 20 61 6e 65 20 68 65 ...n...it's all he...
0040 72 20 66 61 6e 63 79 2c 20 74 68 63 74 3a 20 20 ...Fancy, that...
0050 74 68 65 79 0d 0a 6e 65 76 65 72 20 65 78 65 63 ...they...ver...exc...
0060 75 74 65 73 20 6e 6f 62 6f 64 79 2c 20 79 6f 75 ...ates not only, you...
0070 20 68 6e 6f 77 2a 20 20 43 6f 6d 65 20 6f 6e 21 ...know...Come on!...
0080 27 0d 0a 0d 0a 20 20 60 45 76 65 72 79 62 6f 64 ..."...Everybody...
0090 70 20 71 61 72 73 20 22 63 6f 6d 65 20 6f 6e 21 ...y says "Come on!...
00a0 22 20 68 65 72 65 2c 27 20 74 68 6f 75 6f 68 74 ...here," thought...
00b0 20 41 6e 69 61 65 2c 20 61 73 20 73 68 65 20 77 ...Alice, as she w...
00c0 25 6e 74 0d 0a 73 6e 6f 77 6e 79 20 61 66 74 65 ...ent...slo...ly afte...
00d0 72 20 69 74 3a 20 20 60 49 20 6e 65 76 65 72 20 ...e it...I never...
00e0 77 61 71 20 73 6f 20 6f 72 64 65 72 65 64 20 61 ...was so ordered a...
00f0 62 6f 75 74 20 60 6e 20 61 6e 65 20 64 79 20 66 ...boat in all my l...
0100 69 66 65 2c 0d 0a 6e 65 76 65 72 21 27 0d 0a 0e ...life...no ver!...
0110 0a 20 20 54 68 65 79 20 68 61 64 20 6e 6f 74 20 ...They had not...
0120 27 6f 6e 65 20 6e 63 72 20 62 65 66 6f 72 65 20 ...gone far...before...
```

No.	Time	Source	Destination	Protocol	Length	Info
198	3.439606	192.168.1.222	192.168.1.1	TCP	54	53664 → 53 [ACK] Seq=39 Ack=55 Win=262656 Len=0
199	3.444397	192.168.1.1	192.168.1.222	DNS	144	Standard query response 0x9ab HTTPS gaia.cs.umass.edu SOA unix1.cs.umass.edu
200	3.444950	192.168.1.222	192.168.1.1	TCP	54	53665 → 53 [FIN, ACK] Seq=38 Ack=91 Win=262656 Len=0
201	3.445423	192.168.1.222	128.119.245.12	TCP	66	53669 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
202	3.445658	192.168.1.222	128.119.245.12	TCP	66	53670 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
203	3.449958	192.168.1.1	192.168.1.222	TCP	54	53 → 53665 [FIN, ACK] Seq=91 Ack=39 Win=64256 Len=0
204	3.450034	192.168.1.222	192.168.1.1	TCP	54	53665 → 53 [ACK] Seq=39 Ack=92 Win=262656 Len=0
205	3.489775	128.119.245.12	192.168.1.222	TCP	66	80 → 53670 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
206	3.489775	128.119.245.12	192.168.1.222	TCP	66	80 → 53669 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
207	3.489952	192.168.1.222	128.119.245.12	TCP	54	53670 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
208	3.490020	192.168.1.222	128.119.245.12	TCP	54	53669 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
209	3.512195	192.168.1.1	192.168.1.222	DNS	144	Standard query response 0x5c0a AAAA gaia.cs.umass.edu SOA unix1.cs.umass.edu
210	3.512195	192.168.1.1	192.168.1.222	TCP	54	53 → 53666 [FIN, ACK] Seq=91 Ack=39 Win=64256 Len=0
211	3.512551	192.168.1.222	192.168.1.1	TCP	54	53666 → 53 [RST, ACK] Seq=39 Ack=91 Win=0 Len=0
212	3.560920	192.168.1.222	142.250.9.94	TLSv1.2	148	Application Data
213	3.561176	192.168.1.222	142.250.9.94	TLSv1.2	93	Application Data
214	3.587646	142.250.9.94	192.168.1.222	TCP	54	AA3 → 53623 [ACK] Seq=1 Ack=134 Win=272 Len=0

> Frame 201: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface DeviceNPF_{050:
 > Ethernet II, Src: Intel_4f:f8:0d (48:51:c5:4f:f8:0d), Dst: SagemcomBroad_29:fc:92 (f0:7b:65:29:fc:92)
 > Internet Protocol Version 4, Src: 192.168.1.222, Dst: 128.119.245.12
 > Transmission Control Protocol, Src Port: 53669, Dst Port: 80, Seq: 0, Len: 0
 Source Port: 53669
 Destination Port: 80
 [Stream index: 16]
 > [Conversation completeness: Incomplete, ESTABLISHED (7)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 777519965
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 0
 Acknowledgment number (raw): 0
 1000 = Header Length: 32 bytes (8)
 > Flags: 0x002 (SYN)
 000. = Reserved: Not set

The IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu are **192.168.1.222** and TCP port number used by the client computer is **53669**.

2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

No.	Time	Source	Destination	Protocol	Length	Info
198	3.439606	192.168.1.222	192.168.1.1	TCP	54	53664 → 53 [ACK] Seq=39 Ack=55 Win=262656 Len=0
199	3.444397	192.168.1.1	192.168.1.222	DNS	144	Standard query response 0x9ab HTTPS gaia.cs.umass.edu SOA unix1.cs.umass.edu
200	3.444950	192.168.1.222	192.168.1.1	TCP	54	53665 → 53 [FIN, ACK] Seq=38 Ack=91 Win=262656 Len=0
201	3.445423	192.168.1.222	128.119.245.12	TCP	66	53669 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
202	3.445658	192.168.1.222	128.119.245.12	TCP	66	53670 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
203	3.449958	192.168.1.1	192.168.1.222	TCP	54	53 → 53665 [FIN, ACK] Seq=91 Ack=39 Win=64256 Len=0
204	3.450034	192.168.1.222	192.168.1.1	TCP	54	53665 → 53 [ACK] Seq=39 Ack=92 Win=262656 Len=0
205	3.489775	128.119.245.12	192.168.1.222	TCP	66	80 → 53670 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
206	3.489775	128.119.245.12	192.168.1.222	TCP	66	80 → 53669 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
207	3.489952	192.168.1.222	128.119.245.12	TCP	54	53670 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
208	3.490020	192.168.1.222	128.119.245.12	TCP	54	53669 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
209	3.512195	192.168.1.1	192.168.1.222	DNS	144	Standard query response 0x5c0a AAAA gaia.cs.umass.edu SOA unix1.cs.umass.edu
210	3.512195	192.168.1.1	192.168.1.222	TCP	54	53 → 53666 [FIN, ACK] Seq=91 Ack=39 Win=64256 Len=0
211	3.512551	192.168.1.222	192.168.1.1	TCP	54	53666 → 53 [RST, ACK] Seq=39 Ack=91 Win=0 Len=0
212	3.560920	192.168.1.222	142.250.9.94	TLSv1.2	148	Application Data
213	3.561176	192.168.1.222	142.250.9.94	TLSv1.2	93	Application Data
214	3.587646	142.250.9.94	192.168.1.222	TCP	54	AA3 → 53623 [ACK] Seq=1 Ack=134 Win=272 Len=0

> Frame 201: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface DeviceNPF_{050:
 > Ethernet II, Src: Intel_4f:f8:0d (48:51:c5:4f:f8:0d), Dst: SagemcomBroad_29:fc:92 (f0:7b:65:29:fc:92)
 > Internet Protocol Version 4, Src: 192.168.1.222, Dst: 128.119.245.12
 > Transmission Control Protocol, Src Port: 53669, Dst Port: 80, Seq: 0, Len: 0
 Source Port: 53669
 Destination Port: 80
 [Stream index: 16]
 > [Conversation completeness: Incomplete, ESTABLISHED (7)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 777519965
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 0
 Acknowledgment number (raw): 0
 1000 = Header Length: 32 bytes (8)
 > Flags: 0x002 (SYN)
 000. = Reserved: Not set

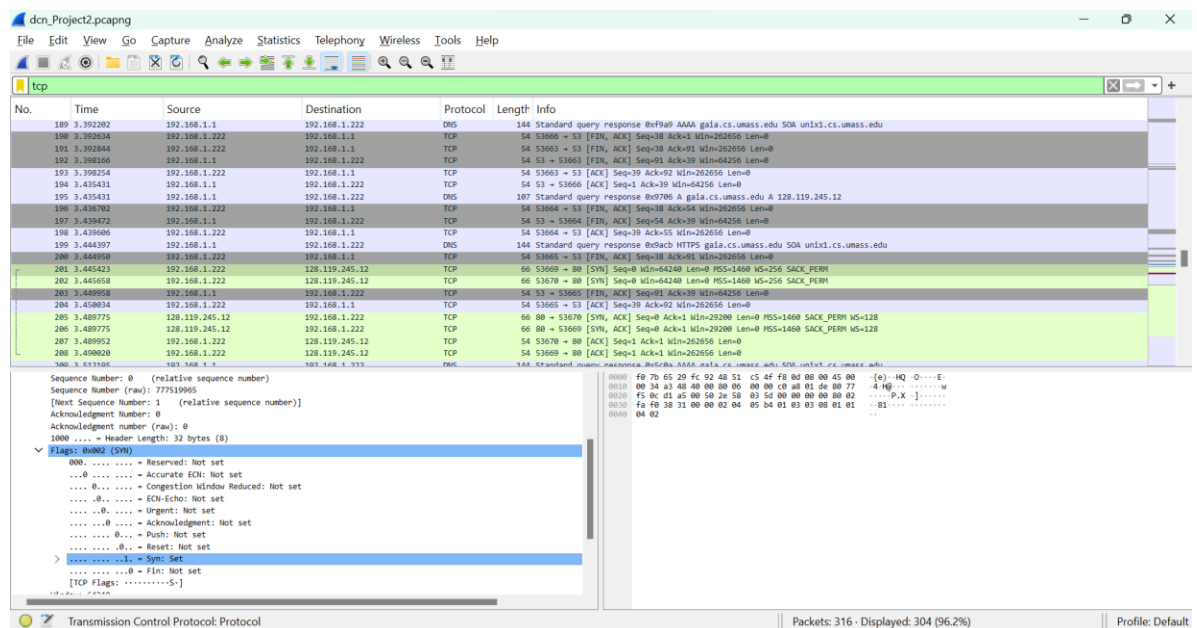
The IP address of gaia.cs.umass.edu is **128.119.245.12** and the it is sending and receiving TCP segments on port number **80**.

1. TCP Basics

Answer the following questions for the TCP segments:

3. What is the *sequence number* of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? (Note: this is the “raw” sequence number carried in the TCP segment itself; it is *NOT* the packet # in the “No.” column in the Wireshark window. Remember there is no such thing as a “packet number” in TCP or UDP; as you know, there *are* sequence numbers in TCP and that’s what we’re after here. Also note that this is not the relative sequence number with respect to the starting sequence number of this TCP session.). What is it in this TCP segment that identifies the segment as a SYN segment?

ANSWER:

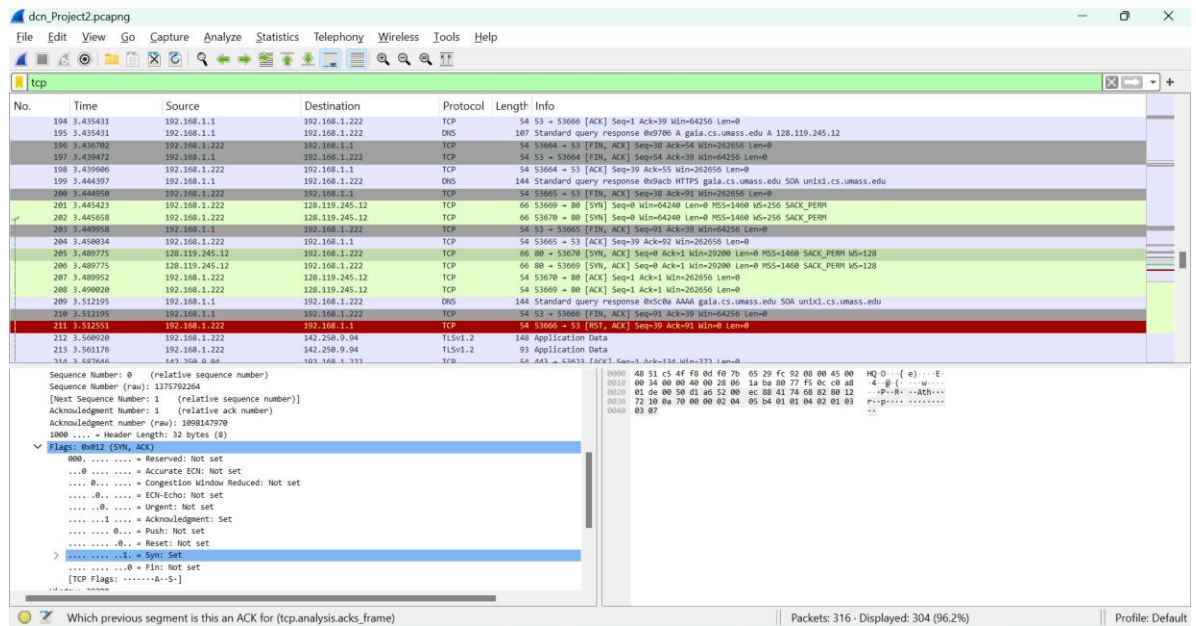


The **sequence number** of the **TCP SYN segment** that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu is **0** in the trace. The SYN flag bit is **set** which identifies the TCP segment as a SYN segment.

Sequence Number(raw): 777519965

4. What is the *sequence number* of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is it in the segment that identifies the segment as a SYNACK segment? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?

ANSWER:



The **sequence number** of the **SYNACK** segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu is **0** in the trace. The **SYN** and **ACK** flag bits are set which identifies the segment as a SYNACK segment.

The **Acknowledgement** number in the SYNACK segment is **1**. gaia.cs.umass.edu determined this value by **adding 1 to the initial sequence number of SYN segment 0**.

Sequence Number(raw): 1375792264

Acknowledgement Number(raw): 1098147970

- What is the sequence number of the TCP segment containing the header of the HTTP POST command? Note that in order to find the POST message header, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with the ASCII text "POST" within its DATA field. How many bytes of data are contained in the payload (data) field of this TCP segment? Did all of the data in the transferred file alice.txt fit into this single segment?

ANSWER:

cdn_Project2.pcapng

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
218	3.587646	142.250.9.94	192.168.1.222	TLSv1.2	85	Application Data
219	3.587646	142.250.9.94	192.168.1.222	TLSv1.2	93	Application Data
220	3.587854	192.168.1.222	142.250.9.94	TCP	54	53623 → 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
221	3.588869	192.168.1.222	142.250.9.94	TLSv1.2	93	Application Data
222	3.589441	192.168.1.222	142.250.9.94	TLSv1.2	89	Application Data
223	3.622388	142.250.9.94	192.168.1.222	TCP	54	443 → 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625707	192.168.1.222	10.18.154.79	TCP	66	53657 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	666	53670 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 → 80 [ACK] Seq=613 Ack=1 Win=262656 Len=11340 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 → 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]

> Internet Protocol Version 4, Src: 192.168.1.222, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 53670, Dst Port: 80, Seq: 1, Ack: 1, Len: 612

Source Port: 53670

Destination Port: 80

[Stream index: 17]

> [Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 612]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 800814790

[Next Sequence Number: 613 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 1375792265

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

0000 = Reserved: Not set

...0 = Accurate ECN: Not set

...0 = Congestion Window Reduced: Not set

...0 = ECH: Echo: Not set

...0 = Urgent: Not set

...1 = Acknowledgment: Set

...1 = Push: Set

...0 = Reset: Not set

...0 = SYN: Not set

...0 = FIN: Not set

[TCP Flags:AP...]

Transmission Control Protocol: Protocol

Packets: 316 · Displayed: 304 (96.2%) · Dropped: 0 (0.0%) · Profile: Default

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
218	3.587646	142.250.9.94	192.168.1.222	TLSv1.2	85	Application Data
219	3.587646	142.250.9.94	192.168.1.222	TLSv1.2	93	Application Data
220	3.587854	192.168.1.222	142.250.9.94	TCP	54	53623 → 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
221	3.588869	192.168.1.222	142.250.9.94	TLSv1.2	93	Application Data
222	3.589441	192.168.1.222	142.250.9.94	TLSv1.2	89	Application Data
223	3.622388	142.250.9.94	192.168.1.222	TCP	54	443 → 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625707	192.168.1.222	10.18.154.79	TCP	66	53657 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	666	53670 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 → 80 [ACK] Seq=613 Ack=1 Win=262656 Len=11340 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 → 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]

Acknowledgment number (raw): 1375792265

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

0000 = Reserved: Not set

...0 = Accurate ECN: Not set

...0 = Congestion Window Reduced: Not set

...0 = ECH: Echo: Not set

...0 = Urgent: Not set

...1 = Acknowledgment: Set

...1 = Push: Set

...0 = Reset: Not set

...0 = SYN: Not set

...0 = FIN: Not set

[TCP Flags:AP...]

Window: 1026

[Calculated window size: 262656]

[Window size scaling factor: 256]

Checksum: 0x3a89 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

> [Timestamps]

> [SEQ/ACK analysis]

TCP payload (612 bytes)

Reassembled PDU in frame: 261

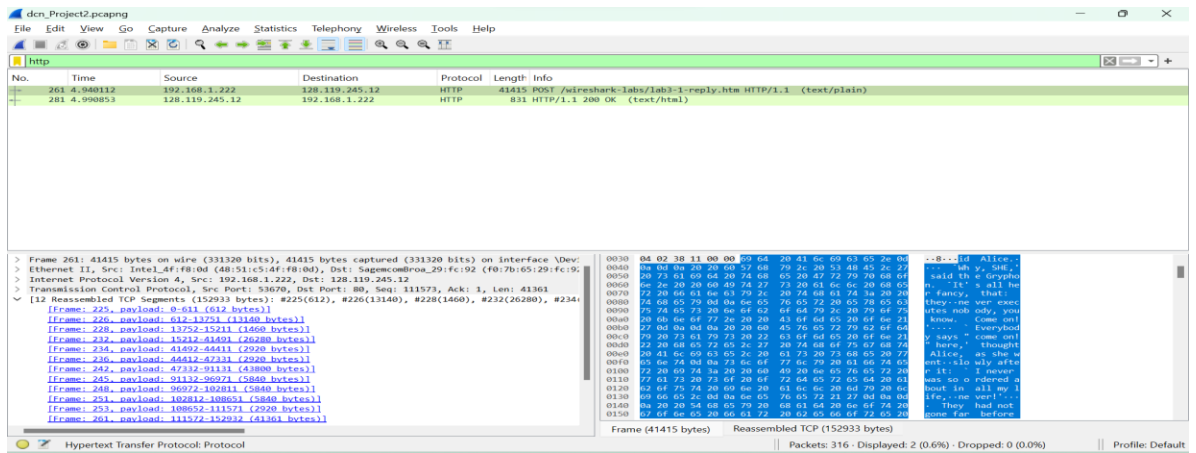
TCP segment data (612 bytes)

Transmission Control Protocol: Protocol

Packets: 316 · Displayed: 304 (96.2%) · Dropped: 0 (0.0%) · Profile: Default

The sequence number of the TCP segment containing the header of the HTTP POST command is 1.

TCP payload: 612 Bytes



No, all of the data in the transferred file alice.txt did not fit into this single segment.

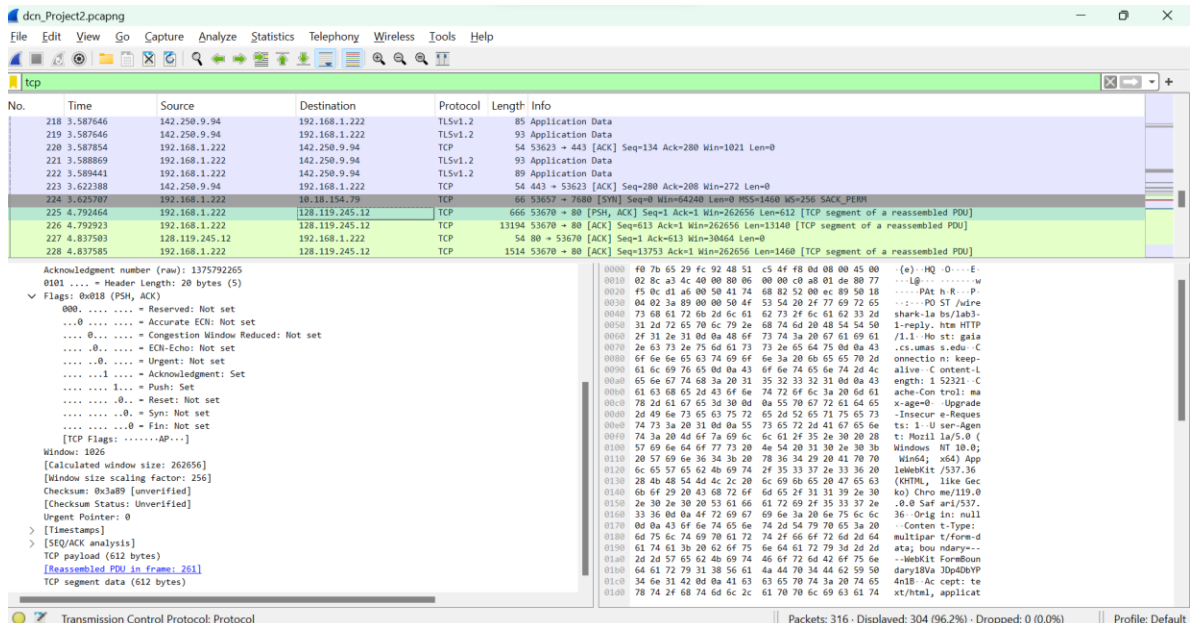
- Consider the TCP segment containing the HTTP “POST” as the first segment in the data transfer part of the TCP connection.

- At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?

ANSWER:

The time, the first segment sent in data transfer part of TCP connection is

4.792464.



- At what time was the ACK for this first data-containing segment received?

ANSWER:

The time, the ACK for this first data-containing segment received was

4.792923.

dcn_Project2.pcapng

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tcp Capture options

No.	Time	Source	Destination	Protocol	Length	Info
218	3.587646	192.168.1.222	192.168.1.222	TLSv1.2	85	Application Data
219	3.587646	192.168.1.222	192.168.1.222	TLSv1.2	93	Application Data
220	3.587854	192.168.1.222	192.168.1.222	TCP	54	53623 → 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
221	3.588869	192.168.1.222	192.168.1.222	TLSv1.2	93	Application Data
222	3.589441	192.168.1.222	192.168.1.222	TLSv1.2	89	Application Data
223	3.622388	192.168.1.222	192.168.1.222	TCP	54	443 → 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625787	192.168.1.222	192.168.1.222	TCP	66	53627 → 7680 [SYN] Seq=0 Win=65536 Len=0 MSS=65536 S=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	666	53670 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 → 80 [ACK] Seq=613 Ack=1 Win=262656 Len=13148 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 → 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]

Acknowledgment number (raw): 1375792265
 0101 ... = Header Length: 20 bytes (5)
 Flags: 0x010 (ACK)
 0000 ... = Reserved: Not set
 ...0 ... = Accurate ECH: Not set
 ...0 ... = Congestion Window Reduced: Not set
 ...0 ... = ECH-Echo: Not set
 ...1 ... = Urgent: Not set
 ...1 ... = Acknowledgment: Set
 ...0 ... = Push: Not set
 ...0 ... = Reset: Not set
 ...0 ... = Syn: Not set
 ...0 ... = Fin: Not set
 [TCP Flags:A....]
 Window: 1026
 [Calculated window size: 262656]
 [Window size scaling factor: 256]
 Checksum: 0x3811 [Unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 [Timestamps]
 [SEQ/ACK analysis]
 TCP payload (13148 bytes)
 [reassembled PDU in frame: 261]
 TCP segment data (13148 bytes)

Transmission Control Protocol: Protocol

Packets: 316 · Displayed: 304 (96.2%) · Dropped: 0 (0.0%)

Profile: Default

- What is the RTT for this first data-containing segment?

ANSWER:

The RTT for this first data-containing segment is **4.792923[ACK received time]- 4.792464[1st data segment transferred time] = 0.000459.**

- What is the RTT value the second data-carrying TCP segment and its ACK?

ANSWER:

dcn_Project2.pcapng

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tcp Capture options

No.	Time	Source	Destination	Protocol	Length	Info
219	3.587646	192.168.1.222	192.168.1.222	TLSv1.2	93	Application Data
220	3.587854	192.168.1.222	192.168.1.222	TCP	54	53623 → 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
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222	3.589441	192.168.1.222	192.168.1.222	TLSv1.2	89	Application Data
223	3.622388	192.168.1.222	192.168.1.222	TCP	54	443 → 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625787	192.168.1.222	192.168.1.222	TCP	66	53627 → 7680 [SYN] Seq=0 Win=65536 Len=0 MSS=65536 S=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	666	53670 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 → 80 [ACK] Seq=613 Ack=1 Win=262656 Len=13148 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 → 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]
229	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=2073 Win=33408 Len=0

Acknowledgment number (raw): 1375792265
 0101 ... = Header Length: 20 bytes (5)
 Flags: 0x010 (ACK)
 0000 ... = Reserved: Not set
 ...0 ... = Accurate ECH: Not set
 ...0 ... = Congestion Window Reduced: Not set
 ...0 ... = ECH-Echo: Not set
 ...1 ... = Urgent: Not set
 ...1 ... = Acknowledgment: Set
 ...0 ... = Push: Not set
 ...0 ... = Reset: Not set
 ...0 ... = Syn: Not set
 ...0 ... = Fin: Not set
 [TCP Flags:A....]
 Window: 1026
 [Calculated window size: 262656]
 [Window size scaling factor: 256]
 Checksum: 0x3811 [Unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 [Timestamps]
 [SEQ/ACK analysis]
 TCP payload (13148 bytes)
 [reassembled PDU in frame: 261]
 TCP segment data (13148 bytes)

Transmission Control Protocol: Protocol

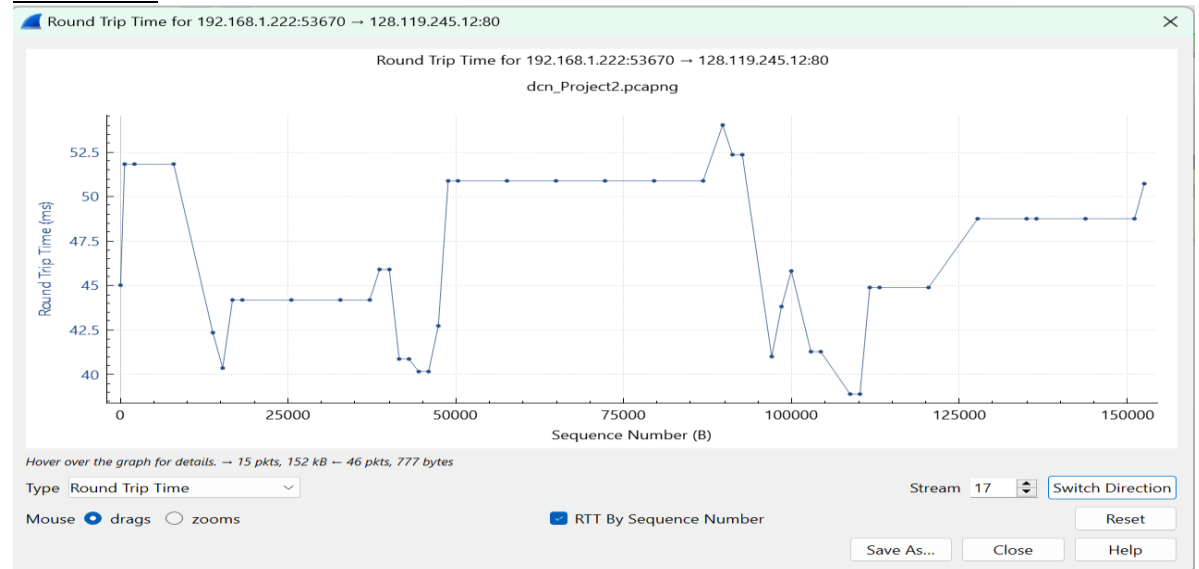
Packets: 316 · Displayed: 304 (96.2%) · Dropped: 0 (0.0%)

Profile: Default

The RTT for the second data-carrying TCP segment is $4.837503 - 4.792923 = 0.04458$.

- Wireshark has a nice feature that allows you to plot the RTT for each of the TCP segments sent. Select a TCP segment in the “listing of captured packets” window that is being sent from the client to the gaia.cs.umass.edu server. Then select: *Statistics->TCP Stream Graph->Round Trip Time Graph*. Plot the RTT and upload a screenshot.

ANSWER:



7. What is the length (header plus payload) of each of the first four data-carrying TCP segments?

ANSWER:

The length (header plus payload) of

1st segment: 612 Bytes

2nd segment: 13140 Bytes

3rd segment: 1460 Bytes

4th segment: 26280 Bytes

No.	Time	Source	Destination	Protocol	Length	Info
219	3.587646	142.250.9.94	192.168.1.222	TLSv1.2	93	Application Data
220	3.587854	192.168.1.222	142.250.9.94	TCP	54	53623 -> 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
221	3.588869	192.168.1.222	142.250.9.94	TLSv1.2	93	Application Data
222	3.589441	192.168.1.222	142.250.9.94	TLSv1.2	89	Application Data
223	3.622388	142.250.9.94	192.168.1.222	TCP	54	443 -> 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625787	192.168.1.222	128.119.245.12	TCP	66	53670 -> 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	66	53670 -> 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 -> 80 [ACK] Seq=613 Ack=1 Win=262656 Len=13140 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 -> 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]
229	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=2073 Win=33408 Len=0
230	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=7913 Win=45056 Len=0
231	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=13753 Win=56704 Len=0
232	4.844824	192.168.1.222	128.119.245.12	TCP	26334	53670 -> 80 [PSH, ACK] Seq=15213 Ack=1 Win=262656 Len=26280 [TCP segment of a reassembled PDU]
233	4.879957	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=15213 Win=59648 Len=0
234	4.880019	192.168.1.222	128.119.245.12	TCP	2974	53670 -> 80 [ACK] Seq=41493 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
235	4.885205	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=16673 Win=62592 Len=0
236	4.885271	192.168.1.222	128.119.245.12	TCP	2974	53670 -> 80 [ACK] Seq=44113 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
237	4.889032	128.119.245.12	192.168.1.222	TCP	54	80 -> 53670 [ACK] Seq=1 Ack=18133 Win=65536 Len=0

8. What is the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments? Does the lack of receiver buffer space ever throttle the sender for these first four data carrying segments?

ANSWER:

The image displays two screenshots of the Wireshark network protocol analyzer, showing the details of a TCP connection to gaia.cs.umass.edu. The top screenshot shows the first four data-carrying TCP segments (seq 6653657 to 6653700) with their advertised receiver window sizes (WS=256, WS=1460, WS=1460, WS=2920). The bottom screenshot shows the same segments with the 'Flags' field expanded, highlighting the 'ACK' flag and the 'Window' field.

Top Screenshot: TCP Segment Details

No.	Time	Source	Destination	Protocol	Length	Info
219	3.587646	142.250.9.94	192.168.1.222	TLSv1.2	93	Application Data
220	3.587854	192.168.1.222	142.250.9.94	TCP	54	53623 → 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
221	3.588869	192.168.1.222	142.250.9.94	TLSv1.2	93	Application Data
222	3.589441	192.168.1.222	142.250.9.94	TLSv1.2	89	Application Data
223	3.622388	142.250.9.94	192.168.1.222	TCP	54	443 → 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625707	192.168.1.222	10.18.154.79	TCP	66	53657 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	666	53670 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 → 80 [ACK] Seq=613 Ack=1 Win=262656 Len=13140 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 → 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]
229	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=2073 Win=33408 Len=0
230	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=7913 Win=45056 Len=0
231	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=13753 Win=56704 Len=0
232	4.844824	192.168.1.222	128.119.245.12	TCP	26334	53670 → 80 [PSH, ACK] Seq=15213 Ack=1 Win=262656 Len=26280 [TCP segment of a reassembled PDU]
233	4.879957	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=15213 Win=59648 Len=0
234	4.880019	192.168.1.222	128.119.245.12	TCP	2974	53670 → 80 [ACK] Seq=41493 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
235	4.885205	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=16073 Win=62592 Len=0
236	4.885271	192.168.1.222	128.119.245.12	TCP	2974	53670 → 80 [ACK] Seq=44113 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
237	4.889032	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=18133 Win=65536 Len=0

Bottom Screenshot: TCP Segment Details (Flags expanded)

Acknowledgment number (raw): 1375792265
0101 ... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
0000 ... = Reserved: Not set
... 0 ... = Accurate ECH: Not set
... 0 ... = Congestion Window Reduced: Not set
... 0 ... = ECH Echo: Not set
... 0 ... = Urgent: Not set
... 1 ... = Acknowledgment: Set
... 0 ... = Push: Not set
... 0 ... = Reset: Not set
... 0 ... = Syn: Not set
... 0 ... = Fin: Not set
[TCP Flags: ...-A-...]
Window: 1026
[Calculated window size: 262656]
[Window size scaling factor: 256]

The top screenshot shows the first four data-carrying TCP segments in the trace. The 'Window size' field is 1026, and the 'Window size scaling factor' is 256. The bottom screenshot shows the same segments, but the 'Window size' field is 26256, and the 'Window size scaling factor' is 102. This indicates that the receiver buffer space advertised to the client is 26256 bytes.

The minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments is 1026.

No, the lack of receiver buffer space doesn't throttle the sender for these first four data carrying segments

- Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?

ANSWER:

No, there are no retransmitted segments in the trace field. The sequence number is increasing throughout the trace.

10. How much data does the receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu? Can you identify cases where the receiver is ACKing every other received segment among these first ten data-carrying segments?

ANSWER:

The receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu has

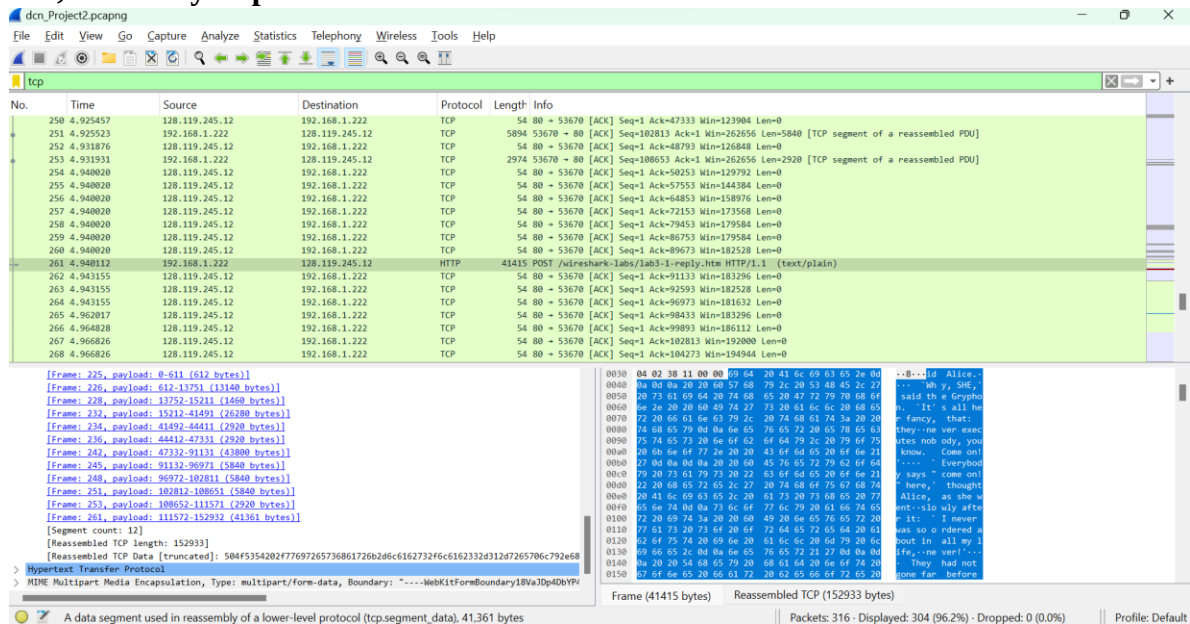
11. What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.

ANSWER:

The total data of alice.txt is 152933Bytes. The transmission time is 4.970854(last ACK) – 4.792464 (1st TCP seg) = 0.17839.

The throughput = total data of file/ transmission time = 152933/0.17839=

857,295.812Bytes per second.



The screenshot shows a Wireshark packet capture of a TCP connection. The packet list displays 26 segments, including 10 data segments and 16 ACK segments. The packet details pane shows the reassembled TCP data (152933 bytes) and the corresponding Hypertext Transfer Protocol (HTTP) data. The packet bytes pane shows the raw data of the selected packet (41415 bytes).

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
250	4.925457	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=47333 Win=123984 Len=0
251	4.925523	192.168.1.222	128.119.245.12	TCP	5894	53670 → 80 [ACK] Seq=108113 Ack=1 Win=262656 Len=5840 [TCP segment of a reassembled PDU]
252	4.931876	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=48793 Win=126848 Len=0
253	4.931931	192.168.1.222	128.119.245.12	TCP	2974	53670 → 80 [ACK] Seq=108653 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
254	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=50253 Win=129792 Len=0
255	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=57553 Win=144384 Len=0
256	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=64853 Win=158976 Len=0
257	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=72153 Win=173568 Len=0
258	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=79453 Win=179584 Len=0
259	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=86753 Win=179584 Len=0
260	4.940820	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=89673 Win=182528 Len=0
261	4.940812	192.168.1.222	128.119.245.12	HTTP	41415	POST /index.html HTTP/1.1 (text/plain)
262	4.943155	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=91133 Win=181296 Len=0
263	4.943155	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=92593 Win=182528 Len=0
264	4.943155	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=96973 Win=181632 Len=0
265	4.962017	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=98433 Win=183296 Len=0
266	4.964828	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=99893 Win=186112 Len=0
267	4.966826	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=102813 Win=192800 Len=0
268	4.966826	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=104273 Win=194944 Len=0

Packet Details:

Frame 225, payload: 0-511 (512 bytes)

Frame 226, payload: 512-13751 (13140 bytes)

Frame 228, payload: 13752-15211 (1460 bytes)

Frame 232, payload: 15212-41491 (26280 bytes)

Frame 234, payload: 41492-44411 (2920 bytes)

Frame 236, payload: 44412-47331 (2920 bytes)

Frame 252, payload: 47332-91131 (43800 bytes)

Frame 245, payload: 91132-96971 (5840 bytes)

Frame 248, payload: 96972-102811 (5840 bytes)

Frame 251, payload: 102812-115271 (2920 bytes)

Frame 261, payload: 115272-152932 (41361 bytes)

Segment count: 12

Reassembled TCP length: 152933

Reassembled TCP Data [truncated]: 504f5354202f7f697265736861726b2d6c6162732f46c6162332d4312d7265706c792d68

Hypertext Transfer Protocol

MIME multipart Media Encapsulation, Type: multipart/form-data, Boundary: "----WebKitFormBoundaryl8VaJOp4DbYV"

Packet Bytes:

Frame (41415 bytes) Reassembled TCP (152933 bytes)

A data segment used in reassembly of a lower-level protocol (tcp.segment_data), 41,361 bytes

Packets: 316 · Displayed: 304 (96.2%) · Dropped: 0 (0.0%) Profile: Default

The top screenshot shows a packet capture of TCP segments. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
220	3.587854	192.168.1.222	142.250.9.94	TCP	54	53623 → 443 [ACK] Seq=134 Ack=280 Win=1021 Len=0
221	3.588869	192.168.1.222	142.250.9.94	TLSv1.2	89	Application Data
222	3.589441	192.168.1.222	142.250.9.94	TLSv1.2	89	Application Data
223	3.622388	142.250.9.94	192.168.1.222	TCP	54	443 → 53623 [ACK] Seq=280 Ack=208 Win=272 Len=0
224	3.625707	192.168.1.222	18.18.154.79	TCP	66	53657 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
225	4.792464	192.168.1.222	128.119.245.12	TCP	666	53670 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=612 [TCP segment of a reassembled PDU]
226	4.792923	192.168.1.222	128.119.245.12	TCP	13194	53670 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=13148 [TCP segment of a reassembled PDU]
227	4.837503	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=613 Win=30464 Len=0
228	4.837585	192.168.1.222	128.119.245.12	TCP	1514	53670 → 80 [ACK] Seq=13753 Ack=1 Win=262656 Len=1460 [TCP segment of a reassembled PDU]
229	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=2073 Win=33488 Len=0
230	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=7913 Win=45056 Len=0
231	4.844764	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=13753 Win=56704 Len=0
232	4.844824	192.168.1.222	128.119.245.12	TCP	26334	53670 → 80 [PSH, ACK] Seq=15213 Ack=1 Win=262656 Len=26280 [TCP segment of a reassembled PDU]
233	4.879957	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=15213 Win=59648 Len=0
234	4.880019	192.168.1.222	128.119.245.12	TCP	2974	53670 → 80 [ACK] Seq=41493 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
235	4.885205	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=16673 Win=62592 Len=0
236	4.885271	192.168.1.222	128.119.245.12	TCP	2974	53670 → 80 [ACK] Seq=44413 Ack=1 Win=262656 Len=2920 [TCP segment of a reassembled PDU]
237	4.889032	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=18133 Win=65536 Len=0
238	4.889032	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=25433 Win=80128 Len=0

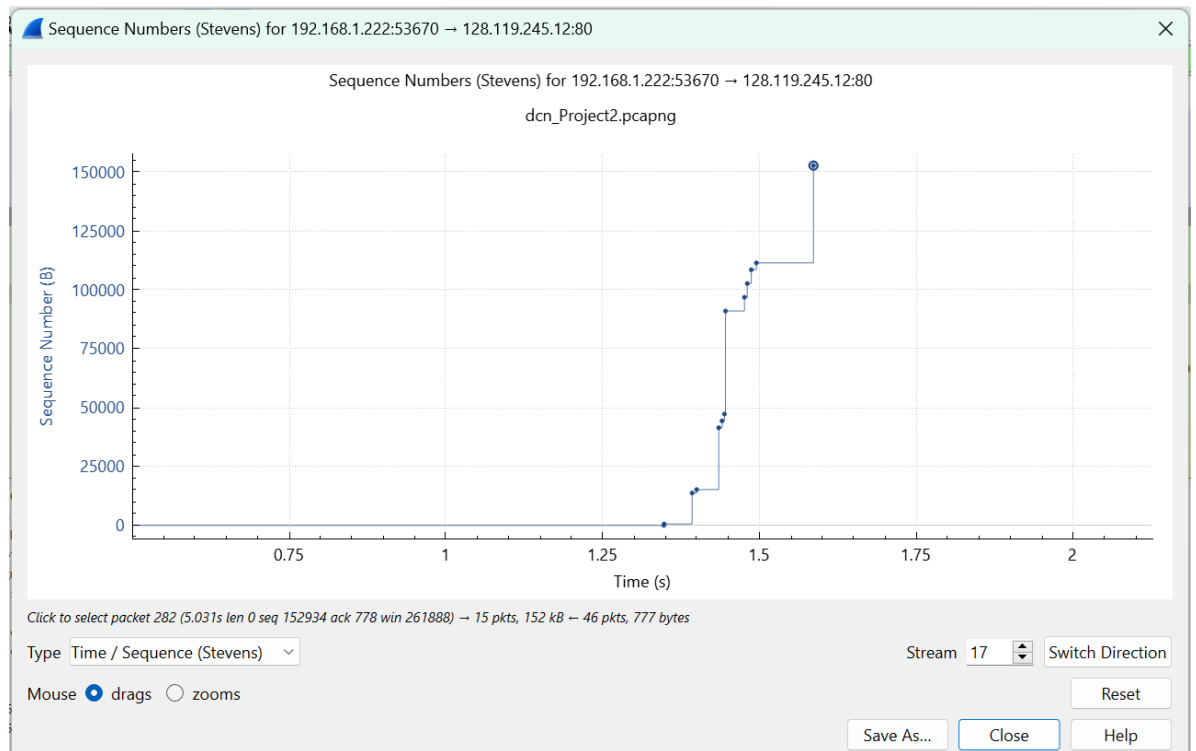
The bottom screenshot shows a packet capture of TCP segments. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
270	4.970854	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=110113 Win=206592 Len=0
271	4.970854	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=111573 Win=209536 Len=0
272	4.985000	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=113033 Win=121800 Len=0
273	4.985000	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=120333 Win=227072 Len=0
274	4.985020	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=127633 Win=241664 Len=0
275	4.988882	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=134933 Win=256256 Len=0
276	4.988882	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=136393 Win=259200 Len=0
277	4.988882	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=146303 Win=273792 Len=0
278	4.988882	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=150993 Win=288384 Len=0
279	4.988882	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=152453 Win=291328 Len=0
280	4.990853	128.119.245.12	192.168.1.222	TCP	54	80 → 53670 [ACK] Seq=1 Ack=152934 Win=294144 Len=0
281	4.990853	128.119.245.12	192.168.1.222	HTTP	831	HTTP/1.1 200 OK (text/html)
282	5.031000	192.168.1.222	128.119.245.12	TCP	54	53670 → 80 [ACK] Seq=152934 Ack=778 Win=261808 Len=0
283	5.532053	2603:6080:4503:3c10:3492::	2603:6080:4503:3c10:3492::	TCP	75	53575 → 443 [ACK] Seq=1 Ack=1 Win=1025 Len=1 [TCP segment of a reassembled PDU]
284	6.505121	2603:6080:4503:3c10:3492::	2603:6080:4503:3c10:3492::	TCP	86	443 → 53575 [ACK] Seq=1 Ack=1 Win=0 Len=0 SLE=1 SRE=2
285	6.875350	192.168.1.222	172.64.41.4	TLSv1.2	110	Application Data
286	6.876344	192.168.1.222	172.64.41.4	TLSv1.2	213	Application Data
287	6.878442	192.168.1.222	172.64.41.4	TLSv1.2	110	Application Data
288	6.879141	192.168.1.222	172.64.41.4	TLSv1.2	213	Application Data

2. TCP congestion control in action

12. Use the *Time-Sequence-Graph(Stevens)* plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Consider the different “fleets” of packets sent in your graph. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase. Figure 6 shows a slightly different view of this data.

ANSWER:



- TCP is in slow start phase and going to congestion avoidance.
13. These “fleets” of segments appear to have some periodicity. What can you say about the period?
- ANSWER:
Periodicity is there.

Submitted by,
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